HAOPENG DENG 鄧浩彭

Guangzhou, China | +86 198-6670-0543 | qianyhp@gmail.com | Google Scholar | Homepage | GitHub

EDUCATION

Guangzhou Maritime University **09/2022-07/2026**

Bachelor of Engineering in *Engineering Management* | School of Future Transportation

Thesis: Attribution Analysis of Digital Retrofitting Effectiveness in Urban Transportation Infrastructure

Research Interests: Mixed-Autonomy Systems; Traffic Flow Modeling and Control; AI for Transportation

ACADEMIC VISITS

Massachusetts Institute of Technology **05/2026-10/2026**

Selected as one of 41 global participants in *MIT-UF-NEU Joint Summer Research Camp* at Prof. Cathy Wu's Lab, collaborating with Ryan Hardesty Lewis on distilling interaction preferences for mixed-autonomy driving.

Tsinghua University & Technical University of Munich **12/2025**

Selected as one of 67 global participants for the "*Smart Sustainable Cities and Transportation*" Winter School, completing 11 modules covering data-driven modeling, mixed-traffic systems, and urban computing.

CONFERENCE PAPERS

[C1] **Deng, H.**, He, S., Wang, D. (2026). FLAT: Smoothing the Rugged Landscape for Learnable, Sample-Efficient Traffic Calibration. Manuscript submitted to *The 38th IEEE Intelligent Vehicles Symposium (IV)*. DOI: 10.48550/arXiv.2610.04337.

[C2] **Deng, H.**, Zheng, F., Xia, X. (2026). SpinFlow: A Physics-Informed Spin Field Framework for Traffic Phase Inference and Transition Detection. *The 29th IEEE International Conference on Intelligent Transportation Systems (ITSC)*. (**Oral Presentation**). DOI: 10.48550/arXiv.2605.23306.

[C3] **Deng, H.**, Zheng, F., Ma, K., et al. (2025). Adaptive Truncated Schatten Norm for Traffic Data Imputation with Complex Missing Patterns. *The 25th COTA International Conference of Transportation Professionals (CICTP)*. American Society of Civil Engineers. (**Best Poster Award**). DOI: 10.1061/9780784486269.009.

[C4] Xia, X., **Deng, H.***, Ye, Y. (2026). Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation. *The 26th CICTP Conference*. *Corresponding author.

JOURNAL ARTICLES

[J1] Xia, X., **Deng, H.**, Tang, J., et al. (2025). Dynamic Traffic Assignment under Mixed Traffic: Modeling, Evolution, and Solution Approaches. *Journal of Transport Science and Engineering*. **(Cover Paper, Oral Presentation at *AIAT 2025*)**. (Chinese). DOI: 10.16544/J.CNKI.CN43-1494/U.20251007001.

[J2] Wang, Y., Li, L., **Deng, H.**, et al. (2026). Sparse Stochastic Optimal Control under Control-Dependent Diffusion and Time-Varying Dynamics. *Journal of Dynamical and Control Systems*. DOI: 10.1007/s10883-026-09784-1.

[J3] **Deng, H.**, Ma, K., Zhan, L., et al. (2026). Structural Equation Modeling of the Relationships among Traffic Safety KAP in Small-Medium Cities. *Journal of Transportation Engineering.* (Chinese). DOI: 10.13986/J.CNKI.JOTE.2026.04.002.

[J4] **Deng, H.**, Tang, J., Zhan, L., et al. (2026). Research on Traffic Governance Optimization in Kaiping City Based on a GRA–FCE Integrated Framework. *Journal of Transportation Engineering.* (Chinese). DOI: 10.13986/J.CNKI.JOTE.2026.04.011.

RESEARCH EXPERIENCE

MIT-UF-NEU Joint Summer Research Camp **05/2026–10/2026**

Title: *Inference-Time Behavior Shaping of Autoregressive Driving Priors for Mixed-Autonomy Traffic Control* | *Research Participant* | *PI: Prof. Cathy Wu* | *Mentor: Ryan Hardesty Lewis*

- Formulated an inference-time behavior-shaping framework for pretrained autoregressive driving models, repeatedly selecting native short-horizon action tokens to steer CAV behavior without updating model weights.
- Built and validated a closed-loop mixed-autonomy platform integrating SMART/CAT-K, MetaDrive, and reactive IDM/MOBIL traffic, with matched counterfactual rollouts and action-execution audits.
- Established repeatable CAV control authority from native model choices across four traffic setups, with all paired continuations preserving directional ordering and 12.7–33.0 m separation after 7 s.
- Demonstrated CAV-to-traffic transmission across six geometry–seed cases, linking follower observations to control and speed changes, and laying groundwork for consequence-aware multi-CAV control.

Guangdong Provincial Special Fund for Science and Technology Innovation Strategy **07/2024-04/2026**
Title: *Cooperative Game-Based Network Traffic Assignment Mechanisms for Urban Transportation Efficiency* | *Project Leader* | *Advisor: Dr. Hu Lei* | *Grant No. PDJH2024A289*

- Authored a systematic review on dynamic traffic assignment (DTA), synthesizing insights from 120+ studies on multi-class vehicle modeling, nonlinear equilibrium evolution, and solution algorithms[J1].
- Developed a VI-based DTA model that unifies UE–SO equilibria into a tunable spectrum, by decoupling CAV penetration into automation levels (AL) and connectivity levels (CL) as evolution variables.
- Formulated and evaluated an adaptive mirror–projection algorithm with active-set warm starts for DUE, DSO, and UE–SO mixed equilibria in a node-based CTM framework under AL–CL heterogeneity.
- Directed team task delegation, milestone tracking, and mentoring of lower year members; coordinated regular progress meetings to align efforts and ensure on-schedule research deliverables.

Guangdong Provincial College Student Innovation & Entrepreneur Training Program **11/2024-02/2026**
Title: *Intelligent Recognition of Urban Traffic Equilibrium States under Air-Ground Collaborative Monitoring* | *Project Co-Leader* | *Advisor: Prof. Xinhai Xia*

- Integrated heterogeneous data sources from roadside sensors, probe vehicles, and UAVs to construct high-dimensional traffic flow tensors, supporting accurate traffic equilibrium state recognition.
- Formulated three representative real-world data loss structures — element-wise, fiber-wise, and mixed — within urban traffic flow tensors, serving as a rigorous testbed for refining data recovery algorithms.
- Developed an adaptive truncated Schatten norm-based tensor completion model to effectively fill gaps in traffic datasets with complex missing patterns, improving MAPE by 10.6% and RMSE by 6.1%[C3].
- Facilitated industry-academia partnership for traffic survey equipment and training, guided junior students in fieldwork to create datasets for analysing traffic flow dynamics and defining equilibrium states.

High-Quality Development Project for Hundred Counties, Thousand Towns, Myriad Villages **07/2024**
Title: *Investigation on Traffic Safety and Planning in Kaiping City* | *Project Leader* | *Advisor: Dr. Hu Lei*

- Collaborated with the Municipal Transportation Bureau to analyse traffic accident and violation data, identifying crash hotspots and associated hazard types, and designing two targeted questionnaires.
- Led a seven-person team to conduct on-site surveys, interviews, inspections, and drone photography, collecting data on traffic flow, vehicle management, pedestrian movement, and infrastructure.
- Utilised SPSS and LLM tools for analysis of survey data and interview transcripts, uncovering key issues and generating actionable insights to inform improvements in urban traffic safety and planning.
- Authored two papers—one on traffic safety behaviour, attitude and cognition relationship using Structural Equation Modeling[J3], and another on the city's traffic governance assessment via Grey Relational and Fuzzy Evaluation[J4]; a synthesized report was later adopted by the Municipal Transportation Bureau.

INTERNSHIP

Guangzhou Schoolplus Software Technology Co., Ltd. **01/2025-05/2025**
Simulation Assistant Engineer | *Mentor: Dr. Wenchao Shen*

- Deployed a multi-agent reinforcement learning module for CAV control in the company's traffic simulator, enhancing its capability to model and visualise V2X-based highway shock-wave propagation.
- Benchmarked the simulator against leading platforms such as SUMO using identical real-world trajectory datasets, diagnosing fidelity gaps and recommending optimisations to enhance competitiveness.
- Surveyed major traffic-related competitions and conferences over the past five years, analysing emerging trends in the field and discovering potential opportunities for industry–academia collaboration.

AWARDS & HONOURS

- IEEE ITSS Travel Grant, The 29th ITSC Conference, ITSS **2026**
- Best Poster Award, The 25th CICTP Conference, COTA **2025**
- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM), COMAP **2025**
- Provincial Third Prize, The 18th "Challenge Cup" National Students' Academic Works Competition **2025**
- Yang Hanlin Scholarship (Top 0.2%, 3/1210), Guangzhou Maritime University **2025**
- Provincial Third Prize, China Undergraduate Mathematical Contest in Modeling **2024**
- Science and Technology Innovation Model Scholarship, Guangzhou Maritime University **2024**

SKILLS & LANGUAGES

Technical Tools: Python, MATLAB, SUMO, LaTeX, Photoshop, Illustrator
Languages: English (IELTS Academic 6.5), Chinese (native, Mandarin & Cantonese)